KANSANSHI TRAFFIC MANAGEMENT STANDARD

KMP_MIN_PRO_002

20.12.2024 _ REV 01

The following document has been reviewed and approved by the following Mine Officials -

APPROVAL LIST				
Role	**Title**	**Name**	**Signature**	**Date**
Document Owner	Production Manager	Bert Scally		14-2-25
Reviewer 1	Chief Mine Planner	Willie Snyman		17/02/25
Reviewer 2	Chief Mining Engineer	George Nzoma		17-02/25
Reviewer 3	Safety Superintendent	Justin Mpesa		14/02/25
Reviewer 4	Training Superintendent	Kenneth .P. Smith		14-02-25
Reviewer 5	Optimization Manager	Kelvin Chitambo		14-02-2025
Reviewer 6	Maintenance Manager	Gregory Mwansa		17.02.2025
Reviewer 7	Health & Safety Manager	Teza Kasengele		18.02.2025
Reviewer 8	Mine Technical Manager	Clayton Reeves		18-02-2025
Approval	Mine Operations Manager	Rees Magrath		18.02.25

FIRST QUANTUM MINERALS | PEOPLE FIRST

1. Purpose

The Kansanshi Mining Traffic Management Plan serves to establish clear direction and an incident free workable approach for all vehicles operating in the mining and associated areas.

The objective of the Kansanshi Mine Operations Traffic Management Plan – Mining (TMP) is to improve road safety, promote safety awareness, improve road traffic conditions and implement improved road traffic management systems in accordance with applicable statutory requirements.

This TMP provides only the minimum requirements for ensuring traffic safety. Specific circumstances and situations could require measures that are beyond those specified in this document. In these instances, specialist professional advice should be sought and a risk based approach adopted.

The objective of the plan is to:

- Improve road safety in the pit;
- Promote safety awareness in the pit and associated areas;
- Improve road traffic conditions; and
- Implement improved road traffic management through better design.

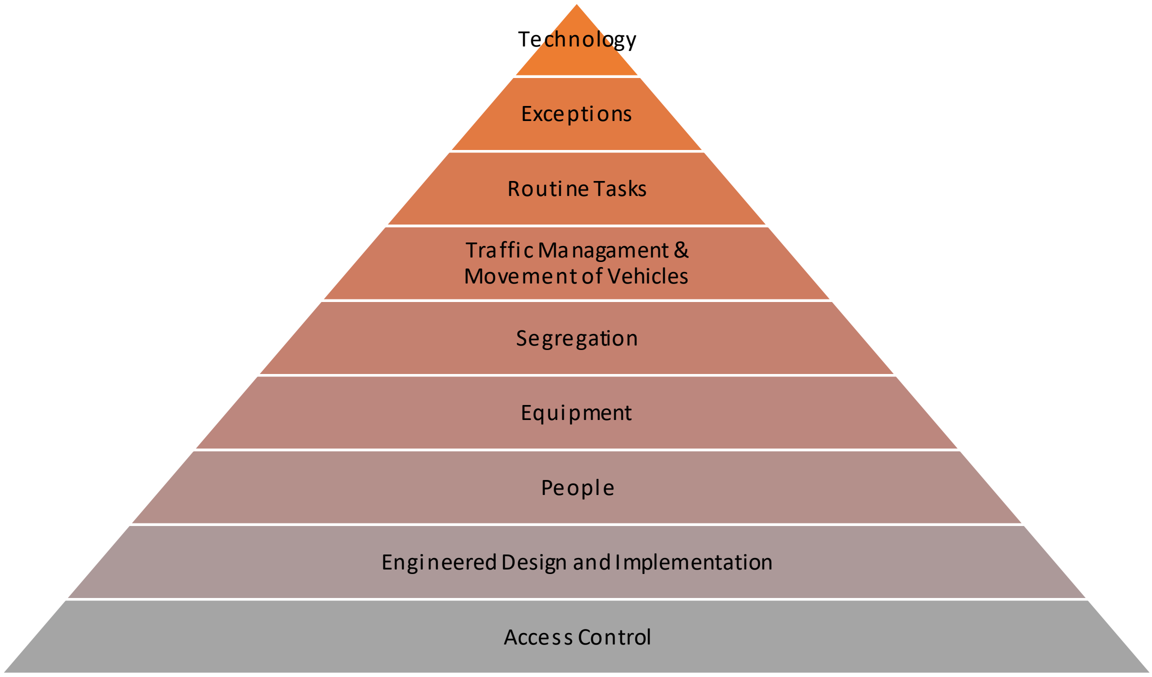
To achieve this, the following methodology is used:

- A comprehensive risk assessment to identify and assess all hazards, potential hazards and conditions that may affect road traffic management;
- Planning road design and setting standards for road construction;
- Training and assessing all personnel in the necessary site procedures required to operate equipment in the pit;
- Promoting safety awareness in the pit at all times;
- Monitoring compliance to Critical Controls/Safe Operating Procedures (C/SWP's) and site road rules.
- Investigating all events and non-conformances associated with operations of vehicles in the pit and associated areas; and
- Complying with TFD's and making the necessary changes as these are amended.

The Traffic Management Plan at Kansanshi Mine is designed to evolve within the framework of the EMERST Vehicle Interaction Control Framework. Its primary objective is to prevent any incidents that could lead to injury or equipment damage in the following three categories:

- Vehicle-to-person interactions
- Vehicle-to-vehicle collisions
- Vehicle-to-equipment or environmental impacts

In doing so, the following document will build as per the below hierarchy.



2. Scope

The requirements of this plan shall apply to all personnel, contractors and visitors to the areas of the Kansanshi Mining Operations jurisdiction managed and operated by the Mining Operations Department.

3. Access Control

The system to manage and control access to and egress from the KMP Kansanshi Mine operation shall ensure the number of Light Vehicles is kept to the minimum required for safe and effective operations. It includes:

- Consistent identification of entrances for road-going vehicles into mining areas that limits access to permit holders only.
- The system to control access into restricted areas where known hazards exist has the following features:
    - Definition of what areas and conditions are considered hazardous and deemed restricted areas.
    - Barriers, signs and delineation used to mark and control access to these areas.
    - Process to obtain entry to these areas.

Access to Restricted Areas

Access to all Restricted Areas will be controlled; limiting access of persons not authorized to enter Restricted Areas. Restricted Areas within the mine site includes, but is not limited to:

- Pit Areas/Mining Areas (Visibly defined by Red “AMA” –Active Mining Area- Cones)
- Delineated haul roads (Visibly defined by Haul Road signage and delineators)
- Designated park-up bays (Visibly defined by White “Light Vehicle Parking” Cones
- Drill and Blast areas (Visibly defined by Yellow and Red Cones dependent on the progression of charging activities)
- Explosive storage facilities (Visibly defined by signage, access control guards and physical access barriers)
- Run of Mine (ROM) areas (Visibly defined by signage at the points of entry)
- Haul Truck Class/Auxiliary Equipment workshop areas (Visibly defined by signage at the points of entry)

- Hazardous materials storage (Visibly defined by signage at the points of entry)
- Other areas as designated by the Production Manager

Restricted areas within the mining lease shall be clearly signposted and/or demarcated as a restricted area in a manner prescribed by the Registered Manager or delegate(s).

Permission to enter these restricted areas shall only be given by the Appointed Manager or delegate(s). Any individual found entering these areas without the permission or approval of an appointed mine official will be subject to sanctions, which may include an MSD breach, disciplinary action, revocation of pit access, and potentially, summary dismissal and removal from the Kansanshi Mine site.

Hazardous Areas

Hazardous areas are those parts of the Kansanshi Mining operation, often limited in size, that present a particular hazard for personnel that are not familiar with specific hazards. Access to these areas shall be restricted to only those personnel that have been deemed competent to operate within these areas.

For hazardous areas where pedestrian access is possible, access shall be through approved pedestrian routes and only after the responsible Mine Coordinator has been notified of their presence.

Hazardous areas at Kansanshi Mine Operations, and covered by this TMP, include, but are not restricted to:

- Explosives yard;
- Magazine
- Sumps
- Charged Blocks
- Drilling Patterns
- Go-Lines
- Hard Parks
- ROM
- Drilling patterns.
- Workshop
- Exploration areas
- Portal areas

All hazardous and high-risk areas shall be identified and clearly marked. Their locations and specific access requirements should be communicated to all relevant personnel. The perimeter of restricted and hazardous areas at Kansanshi should be clearly demarcated by temporary or permanent windrows, barriers, cones, chains, tapes, fence or other suitable means. All entrances to individual areas shall be signed with advisory signage.

Mine Management officials control Vehicle Access (VA) and Individual Access (IA) into the Mining Operational area through a clear process of review and approval. The process and requirements to enter the Mining Jurisdiction for either an individual Pit Permit or a Vehicle Access Sticker are outlined by their respective Pit Access Procedures.

4. Design & Implementation

Road Construction & Maintenance

- Vehicle roadways, haul roads, intersections and hard parks shall be constructed and maintained in accordance with the Kansanshi Mine Haul Road Design Standard.

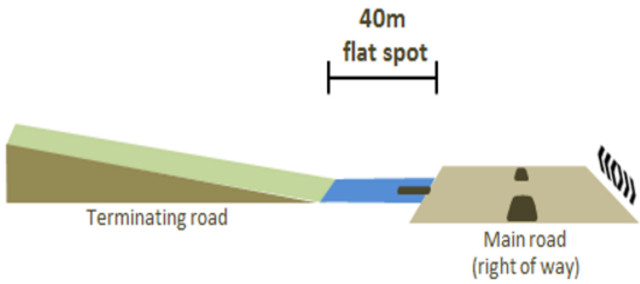
- All Maintenance activities on all site roads – sealed and non-sealed other than risk assessed normal grader operations on mine haul roads – shall be clearly demarcated with signage
- Road intersections shall be designed and constructed to optimize visibility and traffic flow for all vehicles
- Road intersections, where possible, must be constructed at 90 degrees to maximize traffic visibility
- The road shall be maintained by grading, cleaning, sweeping and/or watering when necessary to maintain a smooth running surface, reduce dust generation and provide adequate drainage
- Standardized left side haulage/access road driving direction applies and under no circumstances will the road user be permitted to execute an uncontrolled/ unspotted U-Turn nor is any road user permitted to cross into the on-coming lane of any haul road on any of the Kansanshi mine site traffic routes
- Delineators shall be installed where a roadside hazard has been identified
- Dual traffic roads between the safety windrows should be a minimum of 3 times the width of the widest truck using the road
- Appropriate signage to control traffic flow must be installed.
- All signage installed must comply with standard traffic management control standards.
- The safety berms are constructed to at least ½ of the largest truck wheel diameter, except for intersection corners to allow for maximum visibility. At intersections or switchbacks corners the berm will have a height of 1.0m

Controlled Intersections

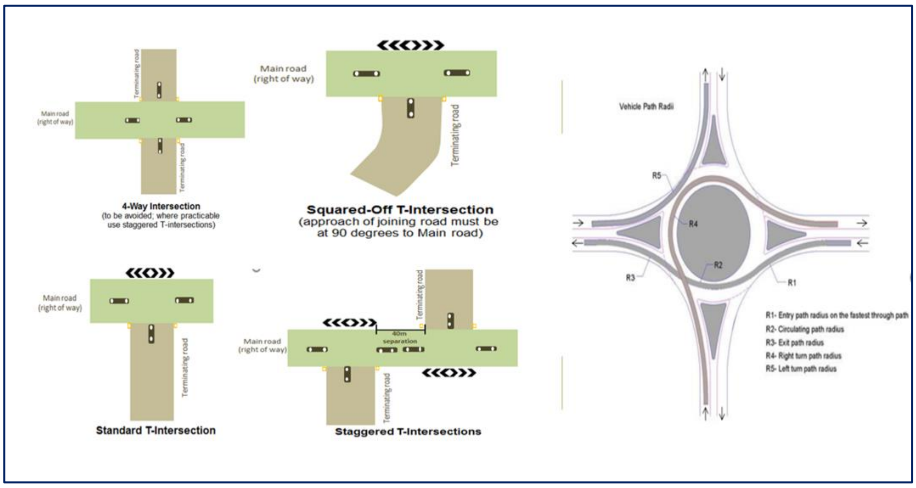
Road intersections create an elevated risk of vehicle-vehicle interaction, and all intersections must be designed to the standards outlined below -

All intersection design standards associated with the KMP Mining Area Jurisdiction are defined within
KA_LH_SOP_060_Stardard for Traffic Management Intersections

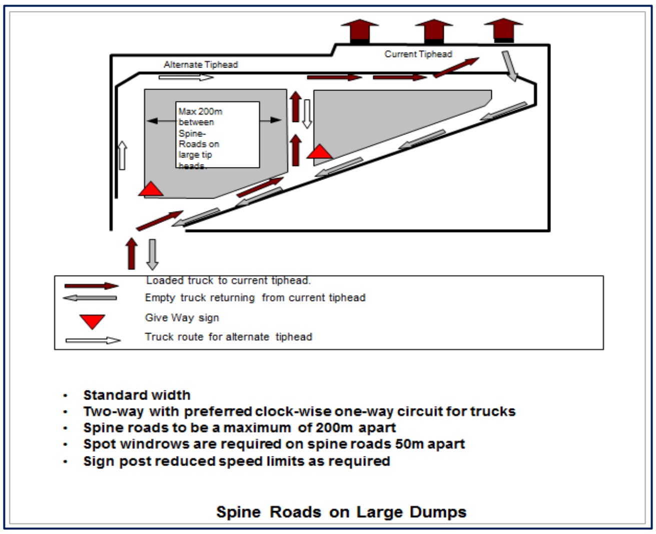
Intersections situated on slopes must have a flat spot leading up to them (no less than 40m) where possible



- An extra 2m in width is to be allowed at intersections to facilitate the placement of splitter islands
- An intersection must not be within 40m of another intersection
- Particular attention should be paid to drainage in intersection areas
- Intersections shall be located to ensure vehicles can stop within the line of sight distance for the designed speed
- Intersections shall be controlled by standardised KMP Road Signage
- Splitter islands at intersections shall be designed to restrict the speed of vehicles crossing the main carriageways. They also shall be designed to provide adequate warning to approaching motorists of the existence of an intersection and ensure that the entering vehicle crosses the main carriageway at a satisfactory angle. This allows for good driver visibility, minimises crossing distances and reduces the potential for high relative speed impacts.
- Intersections shall contain road dividers on each two-way road and the terminating road shall have a “Give Way or Stop” sign in place.
- Below are the standard mine vehicle intersections.



- Below is the preferred waste dumping traffic management design.

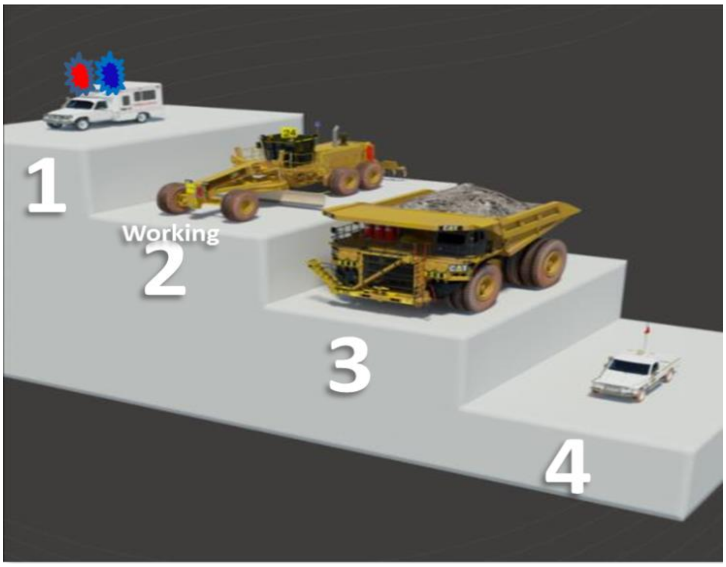


Uncontrolled Intersections

- The following order of priority shall be followed at all uncontrolled intersections:
  - Emergency Vehicles with sirens and flashing lights operating have right of way over all other vehicles;
  - Working Ancillary Equipment has the right of way over all other vehicles, with the exception of Emergency Vehicles;
  - Haul Truck Class and Non-Working Ancillary Equipment have the right of way of all other vehicles, with the exception of Emergency Vehicles and Working Ancillary Equipment; and
  - Light Vehicles do not have right of way over any other vehicle class.
- In addition to this right of way priority, the following give way rules also apply:
  - In Haul Truck Class to Haul Truck Class Equipment encounters, the vehicle on the LEFT has right of way;
  - In Non-Working Ancillary to Non-Working Ancillary Equipment encounters, the vehicle on the LEFT has right of way;
  - In Haul Truck Class to Non-Working Ancillary Equipment encounters, the Haul Truck Class has right of way;

- In Light Vehicle to Light Vehicle encounters, the light vehicle on the RIGHT has right of way;

- All vehicles giving way must come to a complete STOP.



Light Vehicle Segregation

- Kansanshi has implemented a dedicated network of light vehicle (LV) roads, specifically designed to ensure the complete segregation of mobile mining equipment and light vehicles within certain areas of the mine site.
- The segregated roads are clearly signposted, with the intention of guiding LV drivers to use only these designated routes. It is essential that all LV drivers adhere to these designated paths at all times in order to maintain a safe and effective working environment.
- These roads terminate at specific intersections where dual-usage rules of the road resume and light vehicle drivers should apply the highest level of awareness and operational compliance.
- When entering **Active Mining Areas (AMAs),** the rules pertaining to these well-defined areas shall apply and be strictly enforced. See the Active Mining Areas standards in this document.

Cable Tree Usage

Each span of cable trees must be between 16m to 17m wide. A range finder, measuring tape or distometer must be used to confirm the width after cable tree after installation.

- The height at the minimum midpoint of each span must not be less than 11m. Use the electronic measuring pole to confirm the height.
- Cable trees must be installed at all times along main haul roads or access to a loading unit to ensure two - way traffic.
- The installation of two trees for access to a loading unit which should be meant for double side loading should be presided by traffic management control having a give way sign installed for the empty trucks with the trees clearly indicated on the memorandum sent out by the Optimisation team.
- During the relocation of the electric drills or shovels, a single set of trees allowing for single access will only be made under the condition that a Spotter must be in place to control traffic flow at all times, give way sign installed for the empty trucks, trees must be pushed out as soon as the drills

5. People

Driver / Operator – Licences and Competencies

- All drivers/operators of Light Vehicles shall have a current, applicable Zambian driver's licence or acceptable national equivalent for the vehicle they are operating or deemed competent by the Appointed Manager or delegate.
- Prior to any person driving a Light Vehicle or operating any Heavy Mobile Equipment in the pit areas of the mine, the individual shall have been deemed competent and authorised to operate the selected vehicle.
- The competency test shall include both a theory and practical assessment.
- Authorisation of the Pit Permit is done by the Training Superintendent or delegate(s).
- Any person required to operate a Light Vehicle in a Pit Permit area, but who is not the holder of a Pit Permit, shall be escorted and directly supervised by a current Pit Permit holder for the area.
- All persons operating equipment on-site shall be responsible for advising their supervisor immediately of any change in the status of their licences. Appointed Manager Approval shall be obtained for a person to continue operating equipment on site in the case of a change in the status of a licence.
- Operating any vehicle or mobile equipment while under the influence of alcohol or drugs is strictly prohibited. A positive test result while operating vehicles or mobile equipment may lead to instant dismissal.
- It is the sole responsibility of the individual to inform their immediate supervisor about the use of any medication that may cause drowsiness or impair their ability to operate vehicles on site. Failure to do so is equally subject to disciplinary action being instituted.

Heavy Earthmoving Equipment Training

- Operators of all mobile mining and construction equipment must be accompanied by a qualified trainer until deemed competent to individually operate by a Competent Assessor as per site training protocols.
- Authorisation of the Equipment Operator Permit is done by the Training Superintendent or delegate(s).

Driving Permits for Light Vehicles

- Only holders of a valid Light Vehicle driving Pit Permit can operate Light Vehicles in areas where Heavy Earthmoving Equipment are also permitted to operate.
- Two distinct access areas shall be designated within the mine site to control the interaction of Light Vehicles and Heavy Earthmoving Equipment:

- **Pit Permit areas** (designated by the Production Manager and appropriately signed/demarcated)
- **General Permit Areas** – all remaining areas of the mine, not requiring a Pit Permit to drive or are signed restricted access areas

- The Operations Manager or delegate(s) shall review the number of Light Vehicle Permit Holders (General and Pit) on an annual basis to ensure the absolute minimum needed for effective operations.
- Boom gates at access points shall control access to the site.

Obtaining an Open Pit Area Permit

To obtain a Light Vehicle Pit Permit the following requirements must be met.

- Must have a valid Pit Area Induction.
- Meet all General Permit requirements.
- No personnel will be issued with a LV pit permit unless they have experience working in an open pit environment, and have been in possession of a National (or home country) driver's license for over one year from date of original issue.
- Pass a Theory and Practical assessment of the Pit Traffic Rules including mine road layout.
- Pass an on-site Vision Test.
- Be deemed competent to operate a Light Vehicle through assessment of competency (This shall include the use of the vehicle, specifics on the vehicle, and verification that they can operate vehicle).
- Blind Spot Awareness Ride Along - Log a minimum of 2 hours as a passenger on a Haul Truck operating within the pit area. (This registration and timeline must be captured within the Wenco FMS).
- A copy of all permits and competencies will be maintained by Training Department.

<u>Unrestricted LV Pit Permit</u>

- May operate in all mining areas any time of the day.

<u>Restricted Light Vehicle Pit Permit</u>

- Driving in day light hours only
- Restricted LV pit permit holder is only allowed to operate on the ring roads.
- No ROM access,
- No waste dump access
- Able to access the pit floors where LV parking is available
- No workshop access

Responsibilities of Drivers / Operators

- It is every driver's responsibility to ensure his or her safety and that of work colleagues at all times.
- All drivers/operators shall always ensure they demonstrate positive defensive driving behaviors by anticipating the actions of other drivers, driving to conditions and maintaining awareness of changing conditions.
- Always be cautious when driving and expect the unexpected.
- All vehicles must have an appropriate pre-operational safety check prior to use. This will be carried out by a walk-around inspection based on appropriate checklists for the vehicle being used. The operator is to record that the inspection has been carried out. All defects must be reported and where a defect

introduces a risk to the safe operation of that machine, the equipment must not be operated until it is repaired

- All company owned light vehicles shall have a documented pre-operational safety check conducted once in every shift where operational use is continual. At other times before use, the operator shall conduct a walk around inspection
- Where hot seat changeovers are utilized, the driver should pull the vehicle over and complete an inspection after dumping the first load if they are changed into a truck that is loaded. This should be done at the nearest designated location to the dumping point. Unloaded trucks will be changed over in a designated area so the walk around inspection can be completed.
- Must abide by all the site driving rules and regulations as outlined in the policy
- Must observe the national road traffic rules
- Correct PPE should be worn at all times
- Seat belts must be worn at all times
- Adhere to all traffic rules and signs at all times.
- Only approved and appointed permit holders may operate equipment for which their permits are valid
- Maintenance personnel holders of valid move and test permits
- Pre-start inspections must be completed before use.
- No person is permitted to smoke in any vehicles/machine or equipment at any time.

Use of Mobile Phones, Electronic Devices and Books

- While driving, mobile phones are not to be used for general communication.
- Drivers are not permitted to use mobile phones under any circumstance until the vehicle is stationary in a safe location.
- Hands-Free Kits are permitted.
- The use of iPads, Tablets, or other similar portable electronic devices are prohibited while operating vehicles or equipment.
- The use of audio earpieces or headphones are not permitted.
- Only company issued mobile phones are to be used in operational areas unless authorized by your Manager or their delegate as per the procedure.
- The unauthorized use of any electronic device or equipment used whilst operating a vehicle, machine or equipment shall result in disciplinary action, which could include termination of employment.

6. Equipment

Maintenance & Inspection of Vehicles

- Vehicle inspections shall be conducted prior to the operational use on Light Vehicles or Heavy Earthmoving Equipment.
- Regular maintenance/servicing and reporting of maintenance requirements shall form the inspection process of a well-maintained and safe fleet of company owned light vehicles.
- Sub-contractors, visitors and delivery vehicles shall be required to maintain and inspect their own vehicles according to site requirements, and present evidence of these inspections and current vehicle compliance when requested.

Equipment Minimum Requirements

- All vehicles in use at the Mine site must have:
  - Effective operational headlights, taillights and turn indicators.
  - An effective audible warning signal, which can be sounded when the vehicle is about to be moved if clear vision immediately in front of and behind the vehicle is not available to the driver.
  - Adequate seating and operational seat belt for the driver and any passengers.
  - All vehicles must have a braking system capable of effectively stopping and holding that vehicle fully loaded under any conditions of operation when driven in accordance with the manager's instructions.
  - All company owned vehicles and equipment shall be fitted with identification numbers clearly visible from a distance of 50m.
  - All wheeled earth moving machinery and any other vehicle to which it is practicable to fit such a system shall have an independent braking system for use in an emergency in the event of failure of the primary braking system.
  - All vehicles shall be fitted with at least one Dry Chemical Powder Fire extinguisher.

Light Vehicle Requirements

All LVs entering the Pit Area must be equipped with the following operating equipment:

- Fitted with a function C-Track
- Headlights on at all times
- Vehicles must be fitted with a buggy whip that extends not less than 3.5m from ground level.
- A flashing orange light bar must be fitted at the highest point of the vehicle and be visible from all sides of the vehicle.
- A serviceable fire extinguisher.
- A Two Way Radio compatible with the mine requirements.
- A functioning horn.
- A First Aid kit to an acceptable standard.
- High Visibility Reflective Strips
- All company owned vehicles and equipment should be fitted with identification numbers clearly visible from a distance of 50m.
- Any LV entering the pit must be able to engage into 4-wheel drive. When driving in the pit during or after rain, the 4-wheel drive must be engaged depending on the conditions. Engagement of 4-wheel drive is dependent on the conditions and should be used at the driver's discretion.
- Note: Explosive trucks (MMUs) and Magazine Vehicles must comply with the requirements of KA_BL_SWP_009_EXPLOSIVES_TRANSPORTATION_PROCEDURE

Communication

All vehicles used at Kansanshi Mining shall be equipped with a functioning two-way radio compatible with the Kansanshi Mining radio system or have a functioning/charged hand held two-way radio.

Area	Channel
Load/Haul Dispatch	CH 1
Drill & Blast Dispatch	CH 2
Blasting	CH 3
Supervisors	CH 4
Electricians	CH 5

Maintenance	CH 6
Construction	CH 7
Security	CH 8
Safety	CH 9
Broadcast	CH 10
Roads	CH 11
*************	CH 12
Henderson	CH 13
Kansanshi Survey	CH 14
Grade Control	CH 15
RC (Grade Control Drilling)	CH 16

- Personnel shall be made aware of the applicable channels through:
  - Induction;
  - Drivers training;
  - Signage at the entrance to specific areas; and
  - The management of change procedure when applicable.

- During blasting operations, radio silence must be maintained on Channel 1 by those who are not directly involved in blasting. Blasts will be controlled on Channel 1.
- Radio usage is covered by KA_LH_SWP_005_Use of Two-way Radio Procedure
- Two-way radios shall be checked prior to commencement of each shift by contacting any other radio user and asking for feedback.
- When using a two way radio all personnel shall:
  - Speak clearly;
  - Be precise and to the point with the message;
  - Acknowledge a message correctly when received;
  - Identify themselves and the equipment/person they are calling with their vehicle or equipment identification number; and
- Do not use two-way radio communication for idle chatter.
- If a person is unsure of a reply, the message should be repeated until clearly understood.

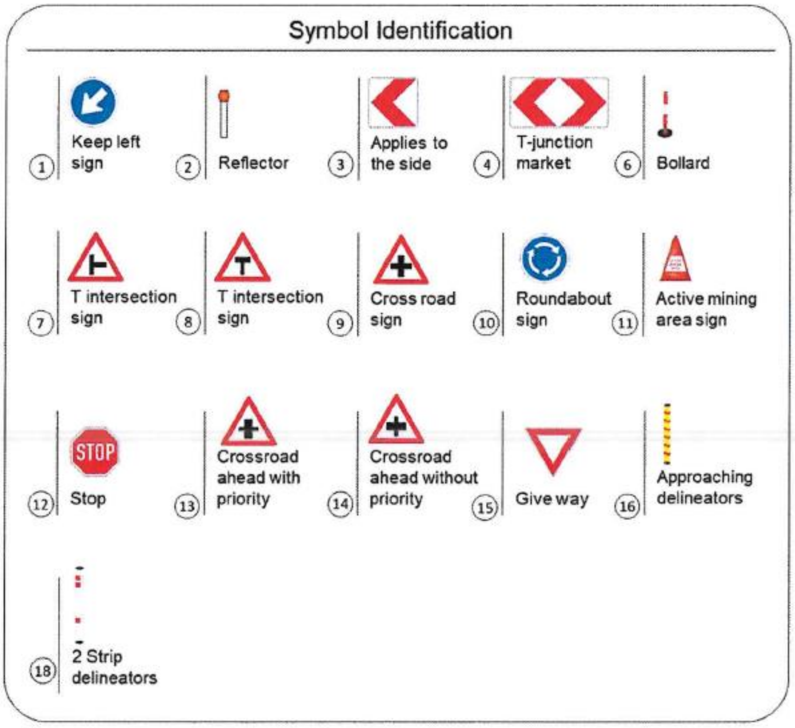
Positive communication is not achieved if there is no response.

- Clicking or visual responses to acknowledge a message shall not be acceptable. A clear answer stating receiving vehicle identification number shall be given.
- Swearing, derogatory comments or offensive language will not be tolerated.

7. Traffic Control

Signs

- The signs used to direct and advise traffic on-site shall conform to the Kansanshi Mine Haul Road Design Standard.
- Custom traffic signs shall only be used when a standard sign cannot convey the desired message. Creation of any new traffic sign shall require approval from the Registered Manager and be supported by a risk assessment.
- Road signs are to be used that conform to Zambian standards. Appropriate signs should be installed at the following locations and as deemed necessary following the completion of a 'Traffic Management Change Request'. The locations are:
    - Intersections
    - Locations where speed limits change
    - Major changes in the grade of roads
    - General road hazards
    - To demarcate Restricted Areas
    - Delineators or guideposts will be installed where a roadside hazard has been identified.



Delineation

Delineation of roads and restricted areas shall be in accordance with the distance and specifications provided in the Kansanshi Mine Haul Road Design Standard.

Lighting

Adequate lighting resources shall be provided on site to service all mandatory areas including:

- Pedestrian crossings
- Permanent walkways
- Go-lines

- Permanent parking and workshop areas
- Active mining areas
- Any other areas as indicated by risk assessment.

Both fixed and portable lighting shall be considered and utilised as appropriate.

8. Movement of Vehicles

Pit Areas

Pit Areas are all areas that form a part of mining activities such as pits, dumps, haul roads, stockpiles, ROMs, crusher pads, drill and blast patterns and any additional areas as sign posted.

Pit Areas shall be designated by the Registered Manager and sign posted and/or demarcated at each mine site, and access to Pit Areas shall be limited to authorized personnel only.

In an emergency, authorized emergency personnel may access Pit Areas.

Vehicle Classifications

All mobile vehicles are classified as either Haul Truck Class, Auxiliary Equipment or Light Vehicles.

- **Haul Truck Class:** Haul Truck Class includes haul trucks, water trucks, low loaders and service trucks (in haul truck configuration).
- **Auxiliary Equipment:** Auxiliary Equipment includes graders, scrapers, wheel dozers, track dozers, drills, production excavators and shovels, production front-end loaders, rock breakers, rollers and small loaders.
- **Light Vehicles:** Light vehicles are all vehicles not already classified as Haul Truck Class or Auxiliary Equipment.

Vehicles Interactions

In all Pit Areas the following rules shall apply as related to vehicle interactions.

Types of Interaction

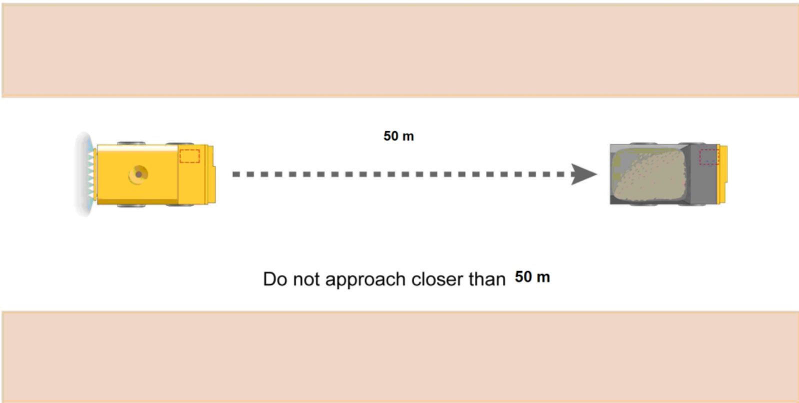
There are **three different types of vehicle interaction** considered in this section.

- **Passing stationary vehicles or equipment**: This refers to one vehicle passing another vehicle that is stationary (e.g., a Light Vehicle passing a broken down haul truck).
- **Overtaking moving vehicles or equipment**: This refers to one vehicle overtaking another moving vehicle travelling in the same direction (e.g. A Light Vehicle overtaking another Light Vehicle).
- **Engagement**: This refers to meetings between Light Vehicles and Haul Truck Class/Auxiliary Equipment outside of designated parking areas (e.g. in pit refuelling, servicing etc.).

General Rules

- General traffic flow follows the normal convention of driving on the left-hand side of the road. In unusual circumstances (e.g. road blockage), the relevant supervisor will advise on any additional procedures or conduct a Risk Assessment (RA)
- All vehicles must respect a minimum 50m following distance from vehicles travelling in the same direction as them (unless in the process of overtaking – described below).
- All vehicles shall adhere to a general 50m exclusion zone around Haul Truck Class and Auxiliary equipment (parked or working), unless there is a hard barrier in between (e.g. a windrow).

- No vehicle may enter the 50m exclusion zone until positive two-way radio contact has been made with the operator of the stationary Haul Truck Class equipment or Auxiliary Equipment and clearance to approach or pass has been granted by the operator who receives the request.
- Note: In addition to these general rules, additional rules must also be observed as outlined in the following sections.
- Under NO circumstances is a Haul Truck to reverse on a haul road, ramp or track unless under direct spotting contact and with Senior Load and Haul Management approval.
- NO vehicles are permitted to perform U-turns on ramps. Only exceptions being working dump truck, graders, dozers loaders building or repairing ramps, graders, loaders, dozers, in an emergency or under JRA or Mine Supervisor direction.
- Under no circumstances is a driver of any vehicle allowed to enter the oncoming lane of any road. This restriction applies throughout the mine site, including but not limited to loading bays, haul roads, ramps, switchbacks, dumps, and ROM pads. The only exception is when a road user is passing stationary vehicles—see below for further details.



Passing Stationary Vehicles or Equipment

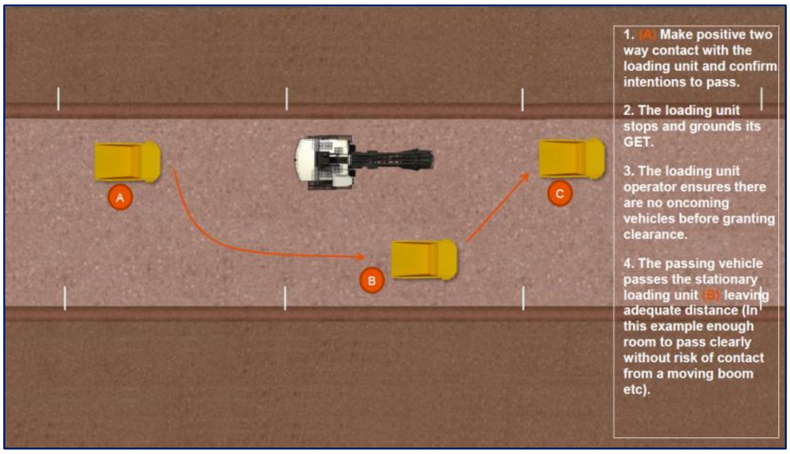
The following rules shall be observed when passing stationary vehicles. Vehicles may only pass stationary Haul Truck Class equipment or Auxiliary Equipment when:

- Positive two-way radio contact has been made with the stationary vehicle's operator/driver and clearance to proceed has been given as per the General Rules described above.
- In this instance, the stationary vehicle's operator/driver must ensure there are no oncoming vehicles or equipment before granting clearance and if relevant ground the equipment's GET.

OR

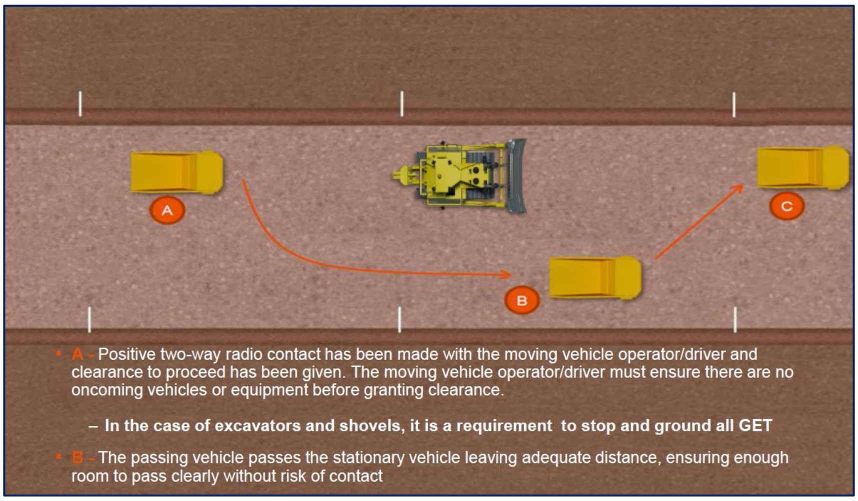
– The vehicle has been authorised or directed to do so by a person in control.

- When passing stationary vehicles, the passing vehicle shall leave an adequate safe clearance distance between the two vehicles and the stationary vehicle must have its GET grounded therefore allowing the passing vehicle to safely pass within the equipment's swing radius (e.g. when passing large excavators on haul roads).
- Passing of light vehicles by other light vehicles must be managed by positive communication where signage or other site processes do not exist.



Overtaking Moving Vehicles or Equipment

- There shall be no overtaking of explosive trucks, emergency vehicles, water trucks, service trucks, light vehicles or haul trucks at any time by any vehicle.
- The following rules shall be observed when overtaking moving vehicles.
- Overtaking moving Haul Truck Class equipment is prohibited at all times.
- Overtaking moving Auxiliary Equipment (except excavators and shovels) or Light Vehicles is only permitted when:
    - Positive two-way radio contact has been made with the moving vehicle's operator/driver and clearance to proceed has been given as per the General Rules described above.
- In this instance, the moving vehicle's operator/driver must ensure there are no oncoming vehicles or equipment before granting clearance.
    - When overtaking moving vehicles or equipment, the speed limit should be observed at all times.
    - When overtaking moving vehicles, the overtaking vehicle shall leave adequate clearance distance between the two vehicles.



Engagement (Approaching Working Equipment)

The following rules shall be observed in Engagements in all areas between Light Vehicles and Haul Truck Class/Auxiliary Equipment.

- Engagement shall be initiated by the Approaching Vehicle driver contacting the Haul Truck Class/Auxiliary Equipment operator as per General Rules detailed above.

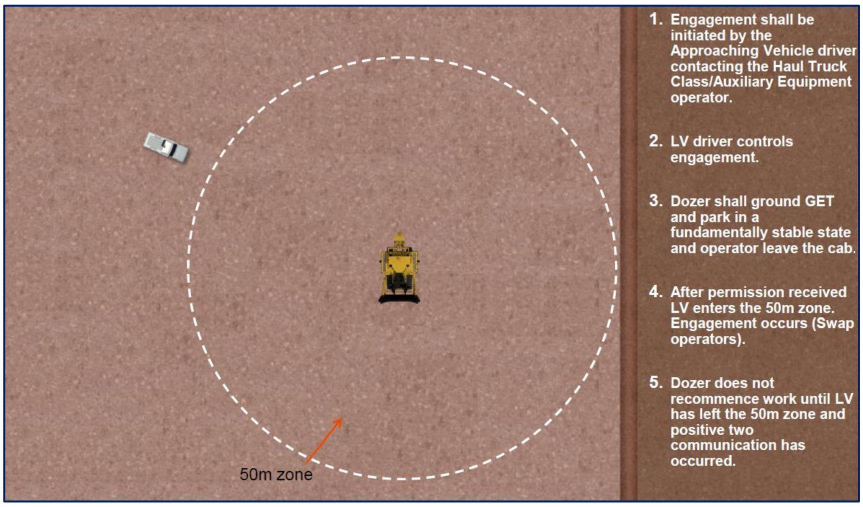
- Where not possible to make positive two-way radio contact because there is no operator on the equipment, the Approaching Vehicle shall obtain permission from the Person in Control before entering the 50m exclusion zone.
- Control of the situation passes to the Approaching Vehicle driver once positive two-way radio contact has been made and the Equipment operator/controller has granted clearance to proceed.
- Control remains with the Approaching Vehicle driver until the driver confirms via positive two-way radio communication that they are clear of the affected exclusion zone as detailed below.
- Before granting permission for the Approaching Vehicle to enter the 50m exclusion zone, the mobile equipment operator shall:
    - Cease all operations, ground any GET, and park in a fundamentally stable state, or park in a designated park up area (Haul Truck Class) if available.
    - After granting permission, the operator shall leave the cab so he/she can be easily seen by the Approaching Vehicle driver.
- When the reason for engagement is concluded, the Approaching Vehicle Driver shall drive out of the 50m exclusion zone and make positive two-way radio contact with the Equipment operator and advise that their vehicle (by stating vehicle number) is clear of the nominated equipment with which the engagement has taken place.
- The Equipment operator should not re-commence operation until he or she has received positive two-way communication that the vehicle involved in the engagement has left the 50m exclusion zone.
- In all instances, it is the Equipment operator's responsibility to ensure the 50m exclusion zone is clear of all Vehicles, personnel and other equipment before moving off.
- Except in special cases (e.g. Breakdown maintenance), Engagement between Light Vehicles and Haul Truck Class/Auxiliary Equipment shall not take place on haul roads.
- Ideally, in the case of rubber tyred auxiliary equipment engagement should take place in safe, level areas with good visibility and sufficient room for the light vehicle to park as per standard safe parking procedures.

Engagement (Approaching Non-Working Equipment)

- Visually ascertain whether the operator is in the cab. If the operator is in the cab, follow above steps for approaching working equipment.
- If the operator is not in the cab, conduct a walk around assessment

Accessing the Waste Dumps & Ore Stockpiles

- If there is a requirement to access a dump area, all drivers should be aware of the 50 metre exclusion zone between any LV and HME (Heavy Mobile Equipment)
- Note - No vehicle is to be driven between the rear of a haul truck and an active tip head point (dumps, crushers or ROM pad).
- Note – all active Waste Dumps and Stockpiles are treated as ACTIVE MINING AREAS at Kansanshi



Escorting

On occasion, vehicles (all classes) with drivers without a Pit Permit will need to access Pit Areas for operational purposes. In these cases:

- Drivers without a Pit Permit that need to access a Pit Area under escort should only be approved by the Production Manager or delegate(s) once the following requirements have been met:
    - Verification that the driver(s) to be escorted has/have a full Zambian Drivers Licence or acceptable national equivalent and any other required licence (i.e. Dangerous goods etc.),
    - Identification of the Pit Area(s) to be accessed, with a plan of the route to be taken,
    - Verification that escorted driver(s) are aware of site emergency procedures and channels,
    - Verification that escorted driver(s) are aware of the route to be taken, the hazards, and the method of the escort.
- All vehicles being escorted shall be equipped with a two-way radio compatible with that of the escort vehicle(s).
- Workgroups in affected areas shall be notified prior to and upon completion of escorting.
- All vehicles forming part of the escort are considered Light Vehicles in respect to the Priority Rules at intersections (except floats/low loaders in specific cases as designated).
- The vehicle being escorted shall remain as close as is practical to the 50m following distance with the escort vehicle or in the case of more than one vehicle being escorted remain as close as practical to the 50m following distance of the vehicle immediately in front within the escorted convoy.
- Light Vehicles under escort shall have a front escort vehicle only.
- Overtaking of escorted vehicles is not permitted at any time.
- Movement of wide and abnormal loads shall be risk-assessed prior to escort.

Breakdowns

- All vehicles that have to stop due to a mechanical breakdown should park up in the nearest safe area and appropriate demarcation shall be put in place.
- Where a vehicle has to stop on a ramp due to an emergency or road obstructions, the operator must turn the wheels fully towards high wall.
- If a vehicle breakdown occurs on a ramp, wheel chocks must be immediately placed to prevent any possibility of vehicle movement before any maintenance, service or repair, and before the operator leaves the cab. The on-shift Supervisor shall organize for a prompt repair or removal of the vehicle. Wheel chocks are not required for LV's but the maintenance team should still secure the vehicle with chocks before attempting to fix the LV
- Should it be likely that the vehicle or mobile equipment will be there for an extended period then a load of material shall be dumped downslope in a fashion that will prevent rolling.
- Repairs to broken down vehicles should only be undertaken at the point of breakdown if there is no risk for the servicing personnel. The area supervisor will advise if works can proceed on the vehicle and organize for the work area to be demarcated and road users to be advised about the repairing activities. Alternatively, the vehicle should be towed away to a safe location for repairs to be undertaken.
- Vehicles shall only be towed with an approved towing device and under maintenance guidance and approval.
- If the break down occurs at night, the parking and/or flashing indicator lights on must be left on and cones must be placed around vehicle.
- Light vehicles responding to a breakdown on a gradient must **always** park on the upslope from any mobile equipment.

In the case of breakdowns in a Pit Area:

- The driver should, where possible, park to the extreme left hand side of the road

- In the case of Light Vehicles, the driver shall park with their park brake engaged, engine switched off, first or reverse gear engaged (or in park for automatic vehicles) and the front wheels turned into the windrow.
- In the case of Haul Truck Class or rubber tried Auxiliary Equipment, the vehicle's wheels should be chocked and front wheels turned into the windrow before exiting the cabin.
- The driver shall activate hazard lights and if the vehicle poses a risk to other vehicles additional controls shall be employed. (i.e.: Hazard triangles).
- The driver shall notify Mine Dispatch and the Area Supervisor that a breakdown has occurred.
- Unless in personal danger, the driver shall remain in the vehicle until assistance arrives.
- If the disabled equipment is located on a haul road or a ramp, the Area or Mine Shift Supervisor notified shall establish roadblocks or windrows at safe distances from the disabled equipment, directing traffic around it.
- Haul Truck Class/Auxiliary Equipment shall be secured as a safety precaution using a mound of dirt tipped immediately in front or behind wheels (as applicable) to prevent it from rolling down a ramp.

Emergency Vehicles

In an emergency:

- Emergency Vehicles shall have right of way at all times.
- When the emergency vehicle is attending an emergency, the entire site shall be informed via radio transmission on all channels that the Emergency Vehicle is active and the site Emergency Management procedure followed.

In-Pit Servicing and Refuelling

The rules for vehicle engagement shall be followed when conducting in-pit servicing or refueling. In addition to these rules:

- Always Park upslope from any mobile equipment.
- Workshop vehicles must park in the designated LV parking bay before approaching trucks that have broken down on the "Hard Park".
- Conduct a walk around inspection of the ME and note any Hazards.
- Ensure Hazards are dealt with before commencing work.
- Only authorized personnel are permitted within the face height of the area being worked and this authorization can only be given by the operator of the equipment working the face.
- The isolation of Haul Truck Class/Auxiliary Equipment during re-fuelling shall be conducted according to site isolation procedures.
- Where possible, in-pit refueling or servicing shall take place in designated areas separated from other traffic.

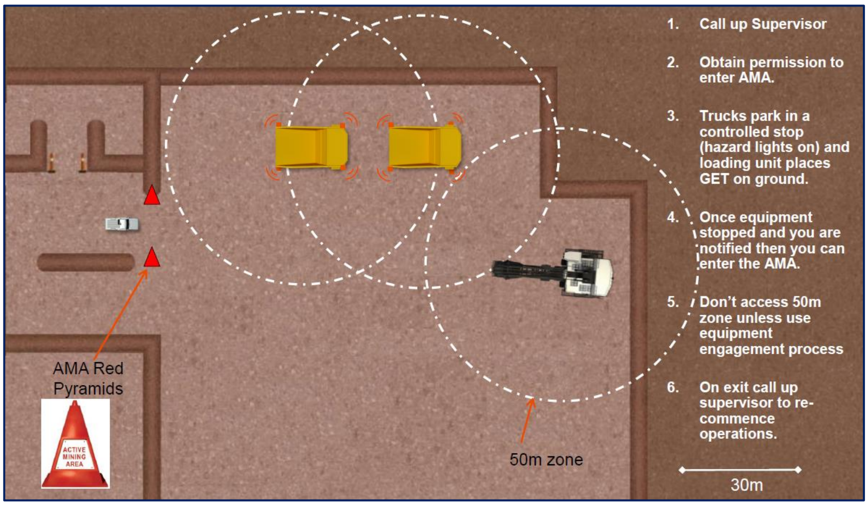
9. Active Mining Areas (AMAs)

An Active Mining Area is in place to delineate a work area from a haul road.

All Active Mining Areas shall be clearly demarcated with Active Mining Area Pyramid Cones (Red).

When a Light Vehicle or pedestrian requires access to an Active Mining Area the following process shall be adhered to-

- The Light Vehicle driver shall request permission to enter the area from the Supervisor or Pit Controller who shall only give authorisation when he/she is satisfied that the requirement for entry is absolutely necessary for the Light Vehicle operator or pedestrian and after all the following provisions have been complied with.
- The Supervisor or Area Supervisor shall ensure that all Heavy Earthmoving Equipment have complied with the following;
    - Are held under a controlled stop within the Active Mining Zone area.
    - Hazard lights turned on.
    - All Loading & Haul Truck Class Equipment and Auxiliary Class Equipment fitted with GET have stopped in a safe location on stable ground and all implements have been lowered to the ground.
- The Supervisor shall confirm by radio with the appropriate equipment (In most cases it will be the Loading Unit Operator) that the area is under a controlled stop.
- The Active Mining Area signs are to remain in place during a controlled stop.
- The Supervisor shall contact the Loading Equipment operator to confirm that the Light vehicle/Pedestrian has left the Active Mining Area and it is clear to resume normal operations.
- When the Active Mining Area is no longer in use the Active Mining Area signage shall be removed and the area either taken control by another work group, made a restricted area (Stockpiles for example), bunded off to prevent access, or converted into use for the next process.
- The use of the 50m exclusion zone for all equipment not directly engaged in the movement of material shall still be used when in an Active Mining Area. Note: Haul trucks lined up behind each other waiting at an excavator shall observe the minimum separation of 10m.
- Access to Active Mining Areas for personnel other than front line operation personnel (Operations Supervisors, Superintendents, Managers, and loading unit operators – e.g. for hot seat change over) should be planned and scheduled during planned equipment downtime (e.g. Fuelling/servicing, Crib Breaks) to minimise equipment interactions.



10. Parking

General

- Design of parking areas and go-lines for all classes of equipment shall be in accordance with Road Design Standard.
- Drive out forward parking is required in all parking bays on the mine where practical.
- All vehicle parking areas should be constructed on a flat area clear of the immediate Active Mining Area.
- No mixing of Light Vehicles & Mobile Equipment shall be permitted within parking areas. Adequate physical separation should be established by erecting appropriate barriers wherever possible.
- Parking areas shall not be used as lay down or storage areas and all parking areas that are no longer required should be removed.
- Parked vehicles shall never be left unattended with the engine running. Vehicle should be parked, hand brake applied and in gear. If the vehicle or equipment is fitted with a turbo timer, then the operator must stay with that vehicle until the timer has shut down the engine.

Parking Mobile Equipment

- To prevent the vehicle from moving in an uncontrolled manner, when parking at a designated parking area, Haul Truck Class and Auxiliary Equipment shall:
- Be parked with the front wheels in the V-drain or rear wheels parked over a raised bumped , and
- Apply the park brake.
- All equipment with moveable attachments and ground engagement tools must be parked with the attachment lowered completely to the ground, and locks and brakes engaged prior to the operator leaving the cab.

Parking Light Vehicles

- Designated delineated LV car parks should be provided in each area to accommodate the expected number of vehicles.
- Parking bays should be located so that approaching traffic is visible to drivers of emerging vehicles.
- Apply park brake – leave in gear.
- Vehicle must not be left unattended with the engine running, or doors/windows open.
- Wheel chocks are not required if a vehicle is parked on the open pit bench, waste dump, ore stockpiles or ROM pad.
- Do not park a vehicle within 15m of a mining face or high wall, or within 50 metres of heavy vehicles if it can be avoided
- Do not park or drive inside a coned off area without authorization; and
- Do not drive over any earth bund or windrow

Parking outside a Designated Parking Area

- No Light Vehicles and Mobile Equipment shall park within the 50m exclusion zone around Mobile Equipment unless they have been granted permission to do so as per the General Rules and in accordance with the Engagement Rules set out in this document.
- In the case that Light Vehicles need to park within 50m of Mobile Equipment (and have been granted permission as per the Engagement Rules):
    - Light Vehicles shall not park or stop on the offside of Haul Truck Class or Auxiliary Equipment within the 50m exclusion zone.
    - Light Vehicles shall not park or stop directly in front of or behind of Mobile Equipment within the 50m exclusion zone.

- Haul Truck Class Equipment and rubber tyred Auxiliary Class Equipment may only park outside a designated parking area if the operator remains in the vehicle (See the Breakdown section for situations where a vehicle is unable to be moved).
- All Light Vehicles and Haul Truck Class/Auxiliary Equipment parked within the Pit Area and not in a designated parking area shall not be parked within 10m of a batter or face; and where practical with the cab side away from the batter or face.

Parking in Inclined Areas

- In normal operating conditions where Mobile Equipment are parked on grades the following applies:
  - The vehicle shall have the engine running, the park brake applied and the operator or driver seated at the controls.
- Where Light Vehicles are parked on grades the following applies:
  - The vehicle shall have the engine running, the park brake applied and the operator or driver seated at the controls; OR
  - The vehicle shall be left in gear with the engine switched off, the park brake applied, and the wheels turned in to the windrow.

Starting Off

- Drivers and operators of all vehicles must check that the vehicle and area required for manoeuvring is clear of obstructions and persons prior to moving off.
- Vehicles must not be driven through any blind area unless the driver has immediately checked that the area is clear or the operator is under the direction of an identified spotter who can safely control the move
- Drivers of heavy vehicles must signal their intentions when starting and moving off. This will be done by using the vehicle horn, and using traffic indicator lights where fitted.
  - *START engine: One (1) blast on the horn and 5 second pause*
  - *Move FORWARD Two (2) blasts on the horn and 5 second pause*
  - *Move in REVERSE Three (3) blasts on horn and 5-second pause.*
- Drivers must drive forward if possible.
- Haul trucks will not be required to use reverse and forward signals when loading from a shovel, excavator, and loader or moving off from dumping.

11. Speed Management

Speed limits should be set to provide a reasonable balance between an acceptable level of control and a driver's perception of the road environment.

Enforcement and education activities should be undertaken to encourage compliance with the speed limits throughout the site.

The below lists the following principles that should be observed when determining speed limits:

- Speed limits shall be capable of being practically and equitably enforced by use of speed zones of adequate length, by limiting speed limit changes and by clarity and frequency of sign posting;
- The speed limit shall not be so low that a significant number of drivers will ignore it.

Speed limits within the Kansanshi Mining areas are:

Light Vehicles	Haul Roads	50 km/h
	LV Access Roads	50 Km/h

	Towing a Trailer or Light Plant	40 Km/h
	LV Ramp Speed	40 km/h
Heavy Mobile Equipment	Ramp Speed – Wet - Empty	15km (under Auto-Retard)
	Ramp Speed – Dry – Empty	30km (under Auto-Retard)
	Ramp Speed – Dry - **Loaded**	15km (under Auto-Retard)
	Haul Roads	40 km/h
	Cornering Speed	15 km/h
	Pit Floors	40 km/h
General	Admin Areas	15 km/h
	ROM Pad	25 km/h
	Fuel Farms	15 km/h
	Workshop	15 km/h

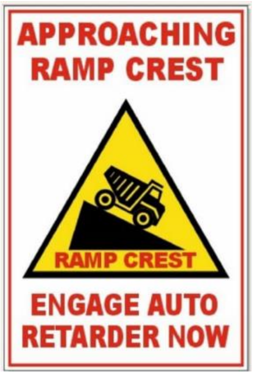
Speed limit signs shall be erected on the left side of the carriageway. Ideally, speed signs should be installed in pairs on both sides of the road when a change of speed zone is required, and as far as practical no other sign should be erected on any post carrying a speed limit sign.

A posted speed limit is the maximum speed for safe driving on a section of road in ideal circumstances.

Drivers shall adjust their speed for the prevailing environment and shall always drive to conditions.

Haul Truck Cresting Speeds

The cresting speed of haulage trucks on ramps is crucial for preventing loss of control. The following key hazardous events can occur in the event of failing to sufficiently apply the critical controls -



- **Retarder Envelope Exceeded** - Occurs when the truck's retarder cannot slow the truck effectively, leading to potential loss of control.
- **Slipping/Sliding Events** - Loss of traction that causes the truck to slide uncontrollably, which is hazardous to both operators and others.

Pre-Crest Speed Control

- Use Auto-Retarders - Always set the auto-retarder to the recommended speed at the crest of the ramp.
- Refer to the Speed Control section of this procedure for exact speed settings based on specific ramp conditions.
- Begin slowing down before reaching the crest of an in-pit, surface, or waste dump ramp.

Retarder Settings and Operation

- Pre-set Speeds - Ensure that the auto-retarder is set at the appropriate speed limit, as outlined in the Speed Control section, before cresting the ramp.
- Verify that the auto-retarder system is functional and properly calibrated.
- Adjust settings as necessary based on environmental and ramp conditions (e.g., wet conditions, dust, fog, etc.)

Training and Awareness

- All equipment operators must be thoroughly trained on:
    - Proper use and settings of the auto-retarder.
    - The importance of decelerating before cresting ramps.
    - Identifying hazardous conditions that may affect traction or braking effectiveness.
    - Operators must continuously monitor the truck's speed and retarder performance during operation, especially at the crest of ramps.

Safety Precautions

- Always ensure that the vehicle is operating within the safety speed limits before, during, and after shift.
- Pay attention to weather conditions (e.g., rain, fog, dust) that can affect traction and braking performance.
- Regularly check the condition of tires and brakes to ensure they are in optimal working order.

12. Additional Considerations

Emergencies

- In the case of an emergency a call "Emergency, Emergency, Emergency" shall be made on:
    - Channel 1 or Channel 10.
    - Telephone number: 0966994588
- The person calling the emergency shall:
    - State their full name;
    - State the location;
    - Describe the nature or type of emergency;
    - Estimate the number of casualties and type of injuries;
    - Assist only if trained to do so;
    - Protect the scene of the emergency and protect other road users from the hazard;
    - Hang up only when told so; and
    - Not use casualty's names over two-way radio.
- In the case of an emergency, all vehicles at KANSANSHI MINING shall park up in the nearest safe location so as not to obstruct emergency traffic on the road and shall wait for instructions from their Mine Coordinator. Drivers and operators shall not leave their vehicles unless instructed.
- Emergency response vehicles with their red and blue flashing lights and sirens on shall have right of way over all vehicles.
- Any injury to a person or damage to equipment must be reported to the appropriate Mine Supervisor as soon as practicable. Failure to do so shall result in disciplinary action.
- Vehicles involved in incidents must not be moved from the incident scene, except in circumstances that may cause risk to personnel, or prevent their safe evacuation.

Overhead Power Lines

- Overhead power lines shall be located, installed and identified in a way that minimises the risk of inadvertent contact, and if practicable, mine roadways shall be designed to minimise crossing underneath overhead power lines.
- The following activities shall not be carried out in the power line corridors which is the area 10m either side of the power line:
    - Drilling, excavating, loading, hauling or dumping.
    - Construction, fabrication, maintenance or storage of buildings, structures, machinery and equipment.
    - Operation of vehicles or machinery with elevating parts that do not afford the required clearance when fully raised.

- All new haul roads shall be designed and located so that they do not pass under overhead power lines where practicable.
- Mobile Earthmoving & Construction Equipment shall only operate along existing or established roads that pass under overhead power lines provided that:
    - Minimum clearance for the movement of vehicles and machinery under and in the vicinity of overhead power lines shall be met.

  - A physical barrier warning device shall be erected along roads passing under the power line with a clearance distance that is less than that of the overhead power lines and provides a safe clearance.
- Warning signs shall be posted on roads under power lines which identify the hazard and the clearance distance.
- Warning signs shall be posted at the entrance of the power line corridor and at a distance that shall allow the vehicle time to stop given the posted speed limit.
- Signage must be installed in a way to alert drivers so as to minimise the risk of inadvertent contact of the structure they are about to go under.

Adverse Weather

- Load and Haul Mine Coordinators shall asses the condition of ramps and haul roads during the shift with special attention during and after rain.
- Any driver or operator that notices slippery or hazardous conditions shall immediately report said conditions to the Load and Haul Supervisors via the two-way radio.
- The Load and Haul Shift Coordinator shall make the call to shut down any ramp or haul road that he/she deems to be unsafe. This shut down can apply to any individual vehicle class or to all vehicle classes at the discretion of the Shift Mine Coordinator.

Pre-Operational Test

- Once the Load and Haul Coordinator deems the Ramp or Haul Road to be safe for operations he/she shall select one sufficiently experienced operator on a selected haul truck to test stopping distances before opening it up to normal operations.
- Before testing a Ramp or Haul Road the Load and Haul Coordinator shall ensure the ramp is free of broken down or parked equipment.
- The Coordinator shall stop the operator at the crest of the ramp or Haul Road Entry and give them specific instructions regarding speed, gear, stopping location, retarder use etc.
- The Coordinator shall then follow the truck from behind and observe the truck for signs of skidding and stopping distances.
- Once satisfied with the condition of the ramp he may then open the area up for normal operations over the two-way radio.
- Dispatch (Control Room) will be the principle and only point of contact for ensuring ramp closure and opening both in the FMS and over the radio channel following approval from the Pit Coordinator.

Deliveries

- Deliveries to any areas within the Kansanshi Mining Area shall be undertaken by applicable boom gates to the following designated routes. Delivery trucks shall enter via the heavy vehicles access in front of the gatehouse where they shall wait for an escort.
- Explosives shall be transported in vehicles complying with the requirements from MSD.
- Any equipment transported in vehicles shall be secured before transporting.
- The use of restraint equipment, where required, shall be used in association with work area procedures.

Cleaning of Vehicles

- All vehicles, including light vehicles that have entered the pit area, shall be kept in a clean and tidy state, and shall be cleaned before any maintenance or repair work is performed.

- When using the automatic wash bays, operators must ensure that there are no other vehicles already in the wash bay. Vehicles must not enter while the sprays are still on, and must not reverse after entering. Operators must drive at a slow speed and keep a straight line.
- All light vehicles are to be kept in a clean condition with adequate visibility. Mud and salt must be removed from the windows and windscreens, and any window cracks should be reported immediately.

Pedestrians Safety

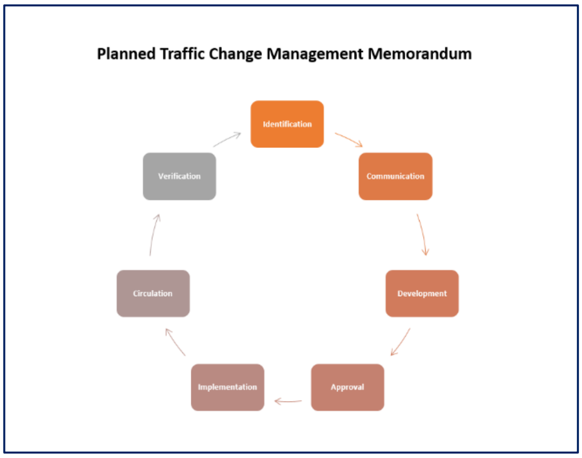
- All personnel are responsible for assessing the risks when they are negotiating traffic. To reduce the opportunity for pedestrian/vehicle interaction the following rules apply:
    - Keep to designated pedestrian paths and crossings where available; and
    - Where there is no designated pedestrian walkway, walk facing oncoming traffic to be aware of movements.
- On Ramps, Roads, Haul Roads, Benches and Pit floors -
    - Pedestrians required to be roadside to conduct work such as signage maintenance, road surveys or incident investigations shall ensure they make positive communication announcing via the two-way radio about their location and intentions prior to exiting their vehicle.
    - Vehicles stopped on the roadside shall be attended to, maintain radio communication and be positioned to the left with the hazard lights on. Pedestrians shall carry a hand held radio to maintain positive communication if the vehicle is left unattended.
    - Cones, delineation and signage shall be used to demarcate the work area and segregate pedestrians from vehicles.
    - The speed limit for passing vehicles with pedestrians on the ground shall be reduced to 20 km/h or the road closed and traffic diverted if passing vehicles pose a risk to pedestrians
    - Personnel on foot must not approach closer than 50m of operating heavy equipment without making radio contact with the operator and other equipment working in the immediate area.
    - Personnel on foot wanting to approach closer than 50m to operating heavy equipment must wait until contact has been made with the operator, the machine has stopped and GET (where applicable) is grounded.
    - Personnel on foot wanting to approach a drill must first seek permission to enter the drill pattern and make direct contact with the drill operator before doing so.

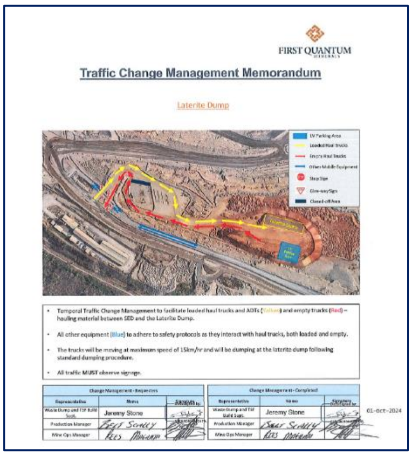
13. Traffic Management Control Changes

The necessity to introduce, remove or adjust existing pit traffic management designs and/or signage will likely occur due to the following events

- Planned - Construction and commissioning of a new ramp or road
- Planned - Blast event and subsequent excavation & haulage resulting in a new ramp or road
- Planned - Closure of a road for fixed infrastructure installations/repairs
- Planned – Reopening of a previously closed ramp or road that requires reactivation after a period of closure
- Unplanned Operational - Reroute of an existing road for approaching drilling patterns
- Unplanned Operational - Identified opportunity for reduced haulage distance resulting in new bench road
- Unplanned Operational - Identification of a traffic hazard and need for urgent adjustment to existing controls.

The below process will apply in the event of any change to Pit Area Traffic Management Controls. Additionally, the standard format for a Traffic Change Management Memorandum will be as follows:





14. Change Management

When a change occurs to this Traffic Management Plan the change shall be communicated to all affected personnel, and management of change shall be initiated.

Only authorised persons or their delegates shall have the authority to make changes to this Traffic Management Plan. Changes must follow the Change Control process defined below.

- The Traffic Management Plan is owned and managed by the Kansanshi Mining Operations Leadership Team.
- The Production Manager retains control of site-specific procedures.
- Any proposed permanent changes to the Traffic Management Plan can be nominated by site Managers Mining Production or nominated delegates to the Mine's Leadership Team for review and approval.

The Mining Operations Management will carry out reviews annually to ensure that the operation are adhering to the Traffic Management Plan.